

Research Proposal: Auditing Fairness in a Clinical Readmission-Risk Model

Team of 3: Smb and groupmates | AI Ethics / Responsible AI course

Research Thesis & Course Connections

Research question: Does a predictive model for 30-day hospital readmission risk exhibit systematic performance disparities (e.g., in false negative rate) across racial and demographic subgroups, and what does this imply for equitable deployment of clinical risk-prediction tools?

Stakes for responsible AI: Clinical risk models increasingly inform real care decisions (follow-up prioritization, discharge planning); an unaudited disparity in error rates translates directly into unequal quality of care, making this a canonical case for algorithmic accountability.

Course concepts grounding this thesis: (1) *disparate impact vs. disparate treatment* as a framework for evaluating algorithmic harm even absent explicit intent; (2) *audit studies as an accountability mechanism* for opaque, deployed models (e.g., the ProPublica COMPAS audit); (3) *fairness metric trade-offs* (e.g., demographic parity vs. equalized odds) and the impossibility of satisfying all simultaneously.

[Replace the three course concepts above with the specific lecture/reading/paper citations assigned in your course, once confirmed.]

Literature & Theoretical Framework

We ground this study in the algorithmic fairness literature applied to healthcare, including Obermeyer et al. (2019, *Science*) on racial bias in a widely used commercial health risk algorithm, and Angwin et al.'s (2016) ProPublica investigation of the COMPAS recidivism tool as a methodological template for auditing opaque risk scores. We also draw on documented racial disparities in pain assessment and treatment (Hoffman et al., 2016, *PNAS*) as broader context for how clinical judgment, and by extension models trained on clinical data, can encode inequitable patterns. As a direct link to this pain-disparity literature, we engineer a **diabetic neuropathy flag** from the diag_1/2/3 ICD-9 codes (chronic nerve-pain complication of diabetes) and examine whether patients with a documented neuropathy diagnosis are treated/flagged differently across racial groups, connecting our audit to the pain-management bias literature rather than only to readmission risk generally. The theoretical framework combines a disparate-impact lens with confusion-matrix-based fairness metrics (true/false positive and negative rates by subgroup).

Audit Study

Dataset: UCI "Diabetes 130-US hospitals" dataset (fully open, no access-gating), containing ~100,000 hospital encounters with race, gender, age, and 30-day readmission outcome fields.

Audit tool: **Aequitas** (dssg/aequitas), an open-source bias and fairness audit toolkit, used via its ready-made Colab notebook workflow to produce group-level confusion-matrix metrics (FPR, FNR, precision) and disparity ratios across race/age/gender subgroups without requiring us to write custom fairness-metric code.

Methodology & Research Design

(1) Clean and preprocess the UCI dataset, selecting a feature set with meaningful clinical/utilization signal (e.g., time_in_hospital, num_medications, number_inpatient/emergency/outpatient, number_diagnoses, A1Cresult, diabetesMed), excluding identifiers and sensitive attributes (race, gender, age) from model inputs; also engineer a binary **neuropathy flag** from diag_1/2/3 ICD-9 codes to link the audit to pain-related complications; (2) train/test split with a fixed random seed; (3) use **Orange** (no-code, drag-and-drop ML tool) to train and compare candidate classifiers (e.g., Logistic Regression vs. Random Forest) via its Test-and-Score and ROC Analysis widgets, justifying our final model choice on accuracy/AUC; (4) run **Aequitas**'s bias audit on the chosen model's predictions, disaggregated by race, gender, age group, and the neuropathy flag, to compute subgroup false negative rate, false positive rate, and disparity ratios against a reference group; (5) interpret disparity ratios against Aequitas's fairness thresholds (based on the 80% rule); (6) discuss which fairness definition (demographic parity vs. equalized odds) is most appropriate for a readmission-risk use case, and why they conflict.

Ethical & Societal Implications

If disparities are found, this raises concrete deployment questions: should a demonstrably unequal model be used at all, and if so, under what monitoring or correction requirements? We will discuss mitigation options (re-weighting, group-specific thresholds, human-in-the-loop review for flagged cases) and the broader pattern this audit sits within — healthcare AI trained on historical data risks encoding, and automating, pre-existing disparities in care.